Non-linear regression with nls

Inference labs use the five-step template: Hypothesis → Visualise → Assumptions → Conduct → Conclude.

Learning objectives

- Fit a parametric non-linear model with `nls()` and report the three parameters.
- Choose starting values that let the optimiser converge.
- Distinguish a GAM's flexible curve from an `nls` model's mechanistic curve.

Prerequisites

Sessions 2 and 4 of this week.

Background

Where a GAM says *fit a smooth and see what shape it takes*, `nls` says *this curve has a known parametric form and I want the parameters*. Michaelis–Menten kinetics, three-parameter logistic dose-response, and exponential decay are all examples of mechanistic models where the parameters have physical meanings: asymptote, half-maximal response, rate, and so on.

Fitting non-linear models is slightly more fragile than fitting linear ones because the optimiser must start somewhere and can get lost. Sensible starting values come from the shape of the data: the asymptote from the highest fitted values, the inflection from where the curve flattens, and so on. `SSlogis` and related self-starting functions estimate these automatically for common parameterisations.

`nls` gives asymptotic standard errors by default. For small samples or poor parameter identifiability, bootstrap or profile-likelihood intervals (`confint(fit)` uses profiles) are more honest.

Setup

```
library(tidyverse)
library(broom)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Hypothesis

Simulate dose-response data and fit a three-parameter logistic.

2. Visualise

```
dose <- rep(c(0.1, 0.3, 1, 3, 10, 30), each = 10)
# true params: asym = 100, xmid = log(3), scal = 0.7 (log-scale)
true_resp <- 100 / (1 + exp(-(log(dose) - log(3)) / 0.7))
resp <- true_resp + rnorm(n, 0, 5)
dat <- tibble(dose, resp)

ggplot(dat, aes(dose, resp)) +
```

```r
  geom_point(alpha = 0.7) +
  scale_x_log10() +
  labs(x = "Dose (log scale)", y = "Response")
```

3. Assumptions

Correct model form, independent normal errors with constant variance on the response scale, and informative starting values.

```r
resid_plot <- tibble(fitted = fitted(fit), resid = resid(fit))
ggplot(resid_plot, aes(fitted, resid)) +
  geom_point() +
  geom_hline(yintercept = 0, linetype = 2, colour = "grey50") +
  labs(x = "Fitted", y = "Residual")
```

4. Conduct

```r
confint(fit)

pred <- tibble(dose = 10 ^ seq(-1, 1.5, length.out = 100))
pred$resp <- predict(fit, newdata = pred)

ggplot(dat, aes(dose, resp)) +
  geom_point(alpha = 0.7) +
  geom_line(data = pred, colour = "steelblue", linewidth = 0.8) +
  scale_x_log10() +
  labs(x = "Dose (log scale)", y = "Response")
```

5. Concluding statement

> A three-parameter logistic fitted on the log-dose scale returned an asymptote of `round(coef(fit)["Asym"], 1)` units and a mid-dose (EC50) at `round(exp(coef(fit)["xmid"]), 2)`. Confidence intervals were obtained via the likelihood profile (see `confint`).

Common pitfalls

- Starting values that put the optimiser in a local minimum.
- Reporting symmetric SE-based intervals when the likelihood is asymmetric; prefer `confint`.
- Over-interpreting parameters when the data do not span the asymptote.

Further reading

- Bates DM, Watts DG. *Nonlinear Regression Analysis and Its Applications.*
- Ritz C, Streibig JC. *Nonlinear Regression with R.*
- Pinheiro JC, Bates DM. *Mixed-Effects Models in S and S-PLUS*, ch. 8.

Session info

```r

```

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)